Compositional Neural Mechanics for Physical Prediction Beyond Visual Agreement
Supplementary Material

Anonymous CVPR submission

1. Scope and experimental map

This supplement specifies the implemented models, numerical methods, experimental accounting, and limitations. The evidence comprises three distinct settings: (i) static composition with oracle physical descriptors, (ii) static inference of 15 continuous unknown descriptors and acquisition of an additional position observation, and (iii) dynamic calibration followed by larger-drop prediction against a native PhysTwin adaptation. Results from these settings are not interchangeable. In particular, the dynamic experiment does not evaluate the static active policy or its uncertainty model.

The central representation is a reusable, conditional energy law. A small shared network maps physical descriptors to coefficients of invariant energy functions. A conventional mechanics solver combines this bulk law with attachment and contact potentials. Scene-specific inference estimates descriptors while the shared network weights remain frozen. Gaussian appearance is attached to material coordinates and transported by the resulting deformation. The system does not learn a direct image-to-force map, reconstruct unknown topology, or learn friction from video.

Units and metric conventions. Coordinates and tool translations are in metres; rotation vectors are in radians; force is in newtons; torque is in newton-metres. Bulk energy density is in pascals, total energy in joules, attachment stiffness in N/m, and normal-contact stiffness density in $\mathrm{N/m^3}$. Static tables report coordinate RMSE. Dynamic benchmark tables report point-vector RMS, which is $\sqrt{3}$ times coordinate RMS for identical samples and weights. Numerical qualification thresholds remain in their originally declared coordinate convention.

What is shared. The bulk and bilateral-interface networks are frozen across objects and mechanical compositions. Reference geometry, material-coordinate fields, prescribed tool poses, gravity, and collision geometry enter the solver explicitly. Gravity is an external body load, rather than an output of the interface network. The dynamic drop study uses the shared bulk law and analytic normal contact; bilateral learned interfaces are evaluated in the static composition study, not covertly inserted into a free-body drop.

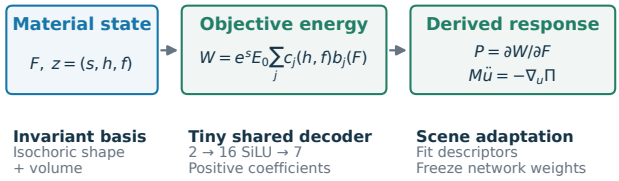


Figure 1. Shared conditional energy components are composed by the mechanics solver. Frozen constitutive weights are distinct from inferred scene descriptors and from transported appearance.

What is conventional. Invariant hyperelasticity, finite elements, sparse Newton solves, penalty contact, consistent mass, Verlet integration, implicit differentiation, and affine covariance transport are established ingredients. Our implementation combines them with shared conditional laws and auditable descriptor inference. We do not claim a new finite-element formulation, proof of polyconvexity, a general collision engine, or calibrated uncertainty.

2. Conditional bulk energy

Let X be a material coordinate, $x(X) = X + u(X)$ the deformation, $F = \partial x/\partial X$, $J = \det F > 0$, and $C = F^\top F$. Define

$$I_1 = \operatorname{tr} C, \qquad I_2 = \tfrac{1}{2}[(\operatorname{tr} C)^2 - \operatorname{tr}(C^2)], \quad (1)$$

$$q_1 = J^{-2/3} I_1 - 3, \qquad q_2 = J^{-4/3} I_2 - 3, \quad (2)$$

$$v(J) = J - \log J - 1. \quad (3)$$

The shared bulk energy is

$$W_\theta(F; s, h, c) = e^s E_0\, a_\theta(h, c)^\top b(F), \qquad E_0 = 12000\,\text{Pa}, \tag{4}$$

with

$$b = [q_1/2, q_2/2, v, q_1^2, q_1^3, \phi_1(q_1), \phi_2(q_1)]^\top, \tag{5}$$

$$a_\theta = \text{softplus}(\text{MLP}_\theta(h, c)) + 10^{-8}, \tag{6}$$

$$\phi_i(q) = \text{softplus}(\alpha_i q) - \log 2 - \alpha_i q/2, \tag{7}$$

$$\alpha_i = \text{softplus}(\rho_i) + 10^{-4}. \tag{8}$$

The coefficient decoder has two inputs, one 16-unit SiLU hidden layer, and seven outputs. Including the two slopes, it contains 169 parameters. The descriptor s is a log common stiffness scale; h controls nonlinear material shape; $c = -1$ and $+1$ designate the two teacher families. The scale enters only as e^s, not through the decoder. Spatial heterogeneity is represented through a material-coordinate descriptor field, rather than a separate network per object.

Objectivity and isotropy follow from the invariants. The reference energy and stress vanish at $F = I$. Stress $P = \partial W/\partial F$ and tangent $\partial P/\partial F$ are evaluated by automatic differentiation. Positive coefficients and reference normalization do not imply global convexity in F, uniqueness of equilibrium, or unconditional stability. The implementation rejects nonpositive determinants and applies additional sampled determinant guards during solves.

3. Constitutive pretraining and interface laws

3.1. Bulk teachers and training

The two analytic teacher families share small-strain moduli but differ at finite strain. With $a = \text{sigmoid}(h)$, $\mu = 4000e^s$ Pa, $K = 20000e^s$ Pa, and $f = .15 + .6a$,

$$W_{\text{MR}} = \frac{\mu}{2}[(1-f)q_1 + fq_2] + Kv, \tag{9}$$

$$W_{\text{Y}} = \mu\left[\tfrac{1}{2}q_1 + (.08 + .3a)q_1^2 + (.01 + .08a)q_1^3\right] + Kv. \tag{10}$$

These laws should not be confused with the separate stable-Neo-Hookean engine-control experiments.

Bulk pretraining uses 18 descriptor combinations: three scales $s \in \{-.5, 0, .5\}$, three shapes $h \in \{-1, 0, 1\}$, and both families. There are 4096 training strain coupons and 512 validation coupons. Deformation gradients are upper triangular with log diagonal entries sampled in $[-.2, .2]$ and upper off-diagonal entries in $[-.4, .4]$. This supplies positive determinants. The objective is

$$\mathcal{L}_W = \text{MSE}(P_\theta/12000, P_*/12000) + .1\,\text{MSE}(W_\theta/12000, W_*/12000). \tag{11}$$

Training uses float64, batch size 256, 6000 fixed Adam steps, initial learning rate .003, and cosine decay. The final iterate is frozen before generating constitutive test coupons. No scene trajectories, reserved-object targets, or rendered future images are used to train this law. The tested interpolation descriptors use scales $\pm.25$ and shapes $\pm.5$; a harder strain test extends the log-diagonal/shear ranges to .35/.7.

3.2. Compliant bilateral attachments

For a tool with reference origin c_b, translation t_b, and rotation R_b, define

$$\delta_b(X) = x(X) - \{c_b + t_b + R_b(X - c_b)\}. \tag{12}$$

The attachment is a bilateral compliant potential, not a unilateral collision constraint. Let $r = \|\delta\|^2/\ell^2$, with $\ell = .002$m. Its learned law is

$$\psi_\theta(\delta; K, \chi, f) = \tfrac{1}{2}K\ell^2\left[r + 2\chi\sum_{j=1}^{7} a_j(\chi, f)\phi_j(r)\right]. \tag{13}$$

The bases comprise r^2 and six functions

$$\phi_j(r) = \text{softplus}(b_j + \alpha_j r) - \text{softplus}(b_j) - \sigma(b_j)\alpha_j r. \tag{14}$$

A two-input, 16-unit Tanh, seven-output decoder produces positive coefficients; six slopes and six offsets are also learned. There are 179 parameters. The outer stiffness factor is K, in N/m. At $\chi = 0$ the interface is exactly quadratic. Tractions and tangents are derivatives of this scalar potential.

The analytic interface teachers use

$$\psi_* = \tfrac{1}{2}K\ell^2\{r + 2\chi[(1-f)r^2 + f(r - \log(1+r))]\}, \tag{15}$$

where $f = 0$ is polynomial and $f = 1$ is saturating in its nonlinear correction. The final wide-range checkpoint warm-starts the earlier small-slip model. Its 16384 coupons comprise equal numbers with radii up to 3ℓ and 16ℓ, and strengths .1, .3, .5, .7, .9. Training uses 6000 fixed Adam steps, batch 512, initial learning rate .003, final rate .0003, and gradient clipping at 1. Force and energy residuals are normalized by $K\ell + \|t_*\|$ and $K\ell^2 + |\psi_*|$, respectively, with energy weight .05. A failed wide-range random initialization remains recorded; it is not substituted into the final method.

3.3. Spatial interface fields

Attachment weights are reference-area weights multiplied by a Gaussian footprint and a smooth boundary taper, then normalized to sum to one. On normalized face coordinates $q = (q_y, q_z)$, the footprint is proportional to

$$\prod_{d=y,z} 4q_d(1-q_d) \exp\left[-\tfrac{1}{2}\sum_{d=y,z}\left(\frac{q_d - c_d}{w_d}\right)^2\right]. \quad (16)$$

Centers c_d and log-widths $\log w_d$ can be unknown continuous descriptors. Normalization makes K a total attachment stiffness rather than a mesh-dependent per-node constant. The weights are differentiable with respect to the centers and widths. Reactions and torques use the same integrated potential and current contact/attachment locations.

Scale ambiguity. For zero independent body load and fixed prescribed tool poses, multiplying all bulk and interface stiffnesses by $\alpha > 0$ gives $\Pi_\alpha(u) = \alpha\Pi(u)$. Consequently the stationary and minimizing sets are unchanged, while reaction forces and tangents scale by α. Position-only data cannot determine this common scale. This is a statement about a specified load model, not global identifiability. Known gravity or known-mass timed dynamics can break the symmetry. A contact term must also be co-scaled for this identity; fixed numerical contact compliance otherwise provides a relative scale.

4. Forward solver, contact and motion

4.1. Finite-element discretization

The displacement field is interpolated with tensor-product quadratic hexahedral basis functions. Each element has 27 nodes, and the structured layout is $(4n, n, n)$ elements. Thus $n = 2, 3, 4, 6, 8$ correspond to 425,1225,2673,8281,18785 nodes, respectively. The parameter n is not the total number of elements per direction. Geometry maps support variations of length, width, depth, taper and arch; arbitrary topology is not inferred.

The discrete potential is

$$\Pi(u) = \sum_q w_q W_\theta(F_q, z_q) + \sum_b \sum_{q\in b} \bar{w}_q \psi_\theta(\delta_q, z_b) + \Pi_c(u) - f_g^\top u \quad (17)$$

Reference-volume and reference-area quadrature are explicit. Static equilibria use sparse Newton iterations with a checked linear residual, descent check and Armijo backtracking. Tangents are energy derivatives. Determinants are checked at quadrature locations and element check points, including nodal locations; these sampled checks are not a global continuous-element positivity certificate. Continuation within a prescribed action improves equilibrium initialization without treating a sequence of unrelated probes as a physical dynamic trajectory.

4.2. Unilateral normal contact

For a known plane or sphere signed distance $g(x)$,

$$\Pi_c = \frac{1}{2} k_c \int_{\Gamma_0} \min(g(x), 0)^2 \, dA_0. \quad (18)$$

The surface points follow the current Q2 deformation, while the integration measure is reference area. The stiffness k_c has units $\mathrm{N/m^3}$. Contact forces and rigid-body reactions are derived from this potential and evaluated at matching positions, including the torque lever arm. The sphere-contact Hessian need not be positive semidefinite because of surface curvature. Penalty contact allows penetration; we report penetration and numerical sensitivity rather than claim exact nonpenetration. Neither friction nor adhesion is included.

The earlier static contact-transfer diagnostic combined both analytic teacher families and the frozen learned laws with this contact potential. Its 16 fine states yielded four passing family/method numerical qualification sets. This is evidence of composition for known descriptors and known collision geometry, not unknown-contact inference or general contact robustness. Tiny-patch sampling failures and earlier resolution failures remain in the development record.

4.3. Actual dynamic integration

Dynamic sequences use a consistent mass matrix,

$$M_{ab} = \int_{\Omega_0} \rho N_a N_b \, dV_0, \qquad M\ddot{u} = -\nabla_u \Pi(u), \quad (19)$$

and velocity Verlet: a half velocity update, position update, force recomputation, and second half velocity update. The mass matrix is factorized once. Free-body scenes remove attachments exactly; they are not stabilized by a weak hidden anchor. Gravity is a known body force. All dynamic videos sample actual integration states rather than interpolating static equilibria.

A separate finite-mass spherical striker stores center, velocity, angular velocity, mass, and spherical inertia. Its normal reaction is the opposite of that on the soft body. The current implementation is not a general rigid-scene collision engine: spherical orientation is immaterial to isotropic normal geometry, and not every rigid–rigid contact pair is supported. Momentum diagnostics use Mv, and angular momentum uses $\sum_a x_a \times (Mv)_a$.

The zero-damping drop benchmark is separate from the exploratory viscous tests. The latter use objective Kelvin–Voigt stress

$$D = \mathrm{sym}(\nabla_X v F^{-1}), \tag{20}$$

$$\sigma_v = 2\eta \,\mathrm{dev}\, D + \zeta \,\mathrm{tr}(D) I, \qquad P_v = J \sigma_v F^{-\top}, \tag{21}$$

with $\eta = 20$, $\zeta = 40$ Pa s and a velocity-dependent explicit-midpoint integrator. Its dissipated power is nonnegative:

$$\dot{D}_{\mathrm{diss}} = \int_{\Omega_0} J[2\eta \| \,\mathrm{dev}\, D\|^2 + \zeta (\mathrm{tr}\, D)^2] dV_0. \tag{22}$$

Both the material and integrator differ from the conservative control, so this is not an isolated causal damping ablation.

4.4. Material-attached appearance

A canonical Gaussian is bound to a material coordinate q with offset $r = \mu_0 - X(q)$. At time t,

$$\mu_t = X(q) + u_t(q) + F_t(q) r, \qquad \Sigma_t = F_t(q) \Sigma_0 F_t(q)^\top. \tag{23}$$

The covariance is refactorized into scale and rotation for rendering. The shared initial appearance checkpoint and opacity filter are used for both methods. Opacity and spherical-harmonic coefficients remain frozen; the returned polar material rotation is not applied to the SH coefficients. Native PhysTwin uses its author transport routine with $K = 16$ neighbors, including its own scale/rotation behavior. Consequently image comparisons include both motion and transport differences. Initial canonical geometry is known, not recovered with COLMAP.

5. Inference and static transfer evidence

5.1. Fifteen continuous unknowns

The static unknown-physics experiment estimates bulk log-scale, shape and heterogeneity amplitude; left/right log-stiffness and nonlinear strength; four support-center coordinates; and four support log-widths. Geometry, material family, interface family, heterogeneity-pattern category, gravity and density are known. Therefore "15 unknown descriptors" does not mean unknown categorical discovery.

For observations y at prescribed poses, the objective combines noise-normalized position residuals and a Gaussian descriptor prior. Implicit derivatives satisfy

$$H \frac{\partial u}{\partial z} = -\frac{\partial R}{\partial z}, \qquad H = \frac{\partial R}{\partial u}. \tag{24}$$

Each parameter direction uses the same equilibrium factorization. Wrench derivatives, when used diagnostically, include both direct parameter dependence and the displacement derivative. The bounded trust-region reflective least-squares optimizer uses Jacobian scaling, a neutral initialization, at most 40 evaluations and 900s per fit. Numerical equilibrium, linear-system and tangent checks precede acceptance of derivatives. The unknown-action study uses position observations only; reaction forces are held predictions, not additional fitting observations.

Method	Composition	New geometry
Matched analytic	.01642	.01590
Neo-Hookean control	.02953	.02982
Shared learned laws	.01811	.01835
Custom direct, stable 6k	3.65036	3.75043
Custom direct, stable 30k	3.08880	3.29332

Table 1. Static oracle-descriptor transfer: displacement coordinate RMSE in mm. Each column contains six objects and four held commands per object. These are not results of unknown-descriptor inference.

5.2. Oracle composition and geometry

The withheld-composition cohort contains six objects. The geometry cohort contains six draws on two unseen geometry parameterizations of the same mapped-cuboid topology, with familiar categorical compositions; these are not six independently new geometry families. Predictions use $n = 4$, references $n = 8$, and 343 material queries. Coordinate RMSE is computed per command, then averaged over four commands and equally over objects. The shared model is close to matched analytic mechanics in this setting (Table 1). Reference resolution was assessed in development; no per-reserved-object continuum certificate is implied.

The direct comparator is a custom DeepSets network, not a published MeshGraphNet implementation. It uses three 128-unit SiLU layers for each node/global encoder, mean and max node pooling, a displacement decoder with three 128-unit SiLU hidden layers and a linear output queried at material points, and a separate 12-output wrench head. Inputs include matched physical descriptors and frozen-law numerical fingerprints. Training uses 54 scenes with six actions each (324 records), and four validation scenes (24 records), with 125 input material nodes and 343 output queries. Its fixed 6k/30k training runs are retained.

The original direct normalization produced severe extrapolation from exactly zero training-feature columns. A separately declared stable-input control replaced only these zero-column scales by public physical units, leaving weights, targets, and nonconstant normalization unchanged. This control was motivated by training/public-input diagnosis, not selecting the best

Shared fitted policy	Pos. mm	Force N	Cov.%
Initial observations	.33949	.09605	97.01
Fixed stretch	.08022	.02921	33.01
Target-task variance	.08022	.02921	33.01
Fixed right bend	.21633	.04278	–
Fixed opposed twist	.08347	.02851	–
Seeded random	.09732	.03429	–

Table 2. Six fresh objects, two held commands per object. Position and force use coordinate RMSE. Cov. is nominal 90% marginal position coverage for the primary local uncertainty approximation. Dashes denote values not tabulated here, not missing prediction attempts.

held prediction. The original results remain archived. A directory-only IO recovery later produced the stable-control artifacts after primary results had been visible; this operational history is disclosed rather than presenting a pristine single run.

5.3. Additional observation and uncertainty

The experiment observes 26 positions with 50μm coordinate noise and predicts two held command responses, using $n = 2$ inference and $n = 4$ references. It fits 48 unique models, of which 41 report optimizer success and seven retain incomplete-optimization forecasts. All 12 selected stretch refits converge. There are 144 complete scored rows across two model types, six policies, six objects and two commands.

The target-task variance policy chooses stretch on all six objects. Thus the improvement after one additional observation is real, but does not demonstrate adaptive superiority over a well-chosen fixed probe. The policy knows the target commands. Point accuracy improves while primary 90% coverage falls sharply. Adding a separately frozen, validation-derived forward-discrepancy mean-square term expands intervals and raises coverage, but this is a supplementary predictive diagnostic: it does not refit likelihoods, change the policy or prove calibration. Model discrepancy at oracle descriptors need not cover inference bias.

6. Fresh dynamic cohort and native comparison

6.1. Data and frozen fitting recipe

Four fresh synthetic worlds use the known $0.12 \times 0.06 \times 0.06$m block and 0.27kg mass. Two worlds per analytic material family are shuffled into opaque identifiers. Private scale and shape are continuously sampled; contact stiffness is sampled log-uniformly between 2×10^7 and 8×10^7 N/m^3. Calibration is a 10mm-clearance drop for 0.20s. Forecasts are 40mm and 80mm drops for 0.30s, with gravity 9.8m/s^2, zero initial velocity and no friction. The reference and our model have no physical damping; native PhysTwin fits drag and dashpot parameters. This tests load-magnitude extrapolation, not new interaction types or geometry.

Both methods receive the same 21 calibration frames at 100Hz, 98 masked surface tracks and known initial geometry/mass. These are oracle 3D tracks with visibility, not tracks reconstructed from raw RGB. Known interior geometry supplies 27 additional initial points for the native particle graph. Private family, material parameters and forecast trajectories are inaccessible to fitting. Forecasts contain 31 saved frames. All predictions are committed before scoring against future targets.

Our dynamic fitting variables are $\xi = (s, h, \log k_c)$ and a binary family candidate. Bounds are $s \in [-1, 1]$, $h \in [-2, 2]$, $k_c \in [10^7, 1.6 \times 10^8]$. The position objective is

$$L(\xi) = \frac{1}{3|\Omega|} \sum_{(t,i)\in\Omega,\ t>0} \|x_\xi(X_i, t) - y_{ti}\|^2. \qquad (25)$$

The 54-candidate catalog spans three scales, three shapes, three contact stiffnesses and two families. Only verified forward trajectories are reused; visibility-weighted losses are recomputed separately for each world. Each family receives a 24-evaluation Powell stage at $n = 2$, followed by 32 evaluations at $n = 3$. The selected prediction uses the lowest-loss saved finite $n = 3$ candidate across both families, not necessarily the optimizer's final iterate. The inference timestep is 50μs. One of 16 optimization stages reports convergence; the other 15 are budget-limited. Nevertheless all eight selected forecasts complete. Completion is not reported as parameter convergence or parameter correctness.

6.2. Native implementation and controls

The native PhysTwin adaptation retains its spring-mass graph, optimizer and prediction routines. Development calibration selected the globally fixed 125-particle extended recipe: at most 100 CMA generations with the author implementation's early stopping, and 1000 epochs. A 729-particle capacity control was also evaluated in development using the same observed tracks and extra known initial geometry. No per-test-world recipe selection is performed. The native model has a different mechanical representation, discretization, contact parameterization and numerical dissipation; it does not receive our analytic teacher law.

The initial native orchestration failed because the child environment did not explicitly satisfy its GPU0 guard. No optimization from that failed attempt was treated as a valid result. An independently reviewed

World	Drop mm	World RMS Ours	World RMS Native	Shape RMS Ours	Shape RMS Native
0	40	.9434	8.8154	.8581	2.4369
0	80	1.1800	19.4244	1.0652	2.8790
1	40	1.1501	8.5297	1.1296	3.2741
1	80	1.4850	17.7201	1.4616	3.7512
2	40	.9680	8.8291	.8928	2.3549
2	80	1.1510	19.3087	1.1050	2.9182
3	40	.8195	Failed	.7987	Failed
3	80	1.0098	Failed	.9623	Failed
Paired six only		1.1462	13.7712	1.0854	2.9357

Table 3. Every fresh forecast attempt. Errors are point-vector RMS in mm, with fixed surface weights; shape removes one proper rigid transformation per frame. Native coverage is6/8, ours8/8. The paired row excludes both methods' world3 scores and is conditional on three completed native fits.

World	Comparison	World	Shape	Pass
0	Calib. $n4/n6$	.2235	.1813	No
0	80mm $n4/n6$	.6162	.4198	No
1	Calib. $n4/n6$	.2028	.1897	No
1	80mm $n4/n6$	.7275	.4958	No
0	80mm $n4$ time	.00987	.00985	Yes
0	80mm $n6$ time	.02032	.02004	Yes
1	80mm $n4$ time	.01876	.01755	Yes
1	80mm $n6$ time	.02613	.02547	Yes

Table 4. All eight fixed-subset sensitivity comparisons, coordinate RMS in mm. Time compares 25 to 12.5μs. Common $6 \times 17 \times 17$ face samples and all saved frames are used. The world/shape thresholds are 0.300/0.150mm; all four spatial comparisons fail and all four temporal comparisons pass.

World	Reference	World RMS Ours	World RMS Native	Shape RMS Ours	Shape RMS Native
0	$n4$	1.180	19.424	1.065	2.879
0	$n6$	1.690	18.779	1.059	2.858
0	$n6$, half step	1.697	18.771	1.059	2.857
1	$n4$	1.485	17.720	1.462	3.751
1	$n6$	1.660	17.005	1.444	3.658
1	$n6$, half step	1.659	17.011	1.446	3.659

Table 5. Explicitly post-hoc secondary robustness check on the two already fixed 80mm cases. Frozen predictions are unchanged. Point-vector RMS in mm. The ranking persists, but the spatial reference failure is not thereby repaired.

operational recovery used separate output directories and explicit environment settings, preserving the original failures, model recipe and frozen public inputs. The recovery completes three world fits and six forecasts. World 3 remains unavailable; it is not replaced by a new world or assigned an invented error.

The result is a controlled synthetic comparison to an adapted native method, not an official PhysTwin benchmark score or a broad SOTA claim. Analytic reference generation is closer to our model class than to a spring network. That advantage is part of the experimental design and limits generalization of the ranking. The native representation's scale and energy dissipation cannot be inferred from our continuum parameters alone.

7. Reference sensitivity, rendering and failures

7.1. Finite-resolution reference assessment

All 20 declared reference jobs complete and pass their individual trajectory checks. The primary generator is $n = 4$, $\Delta t = 25\mu$s. The fixed subset consists of one world from each family (worlds 0 and1), each with $n = 6$ calibration and 80mm forecast companions and half-step80mm runs at both resolutions. These are finite-resolution analytic references, not continuum ground truth.

The sampling norm in this qualification is a uniform face-sample coordinate norm, not an exact mass integral and not the 98-point benchmark weighting. Failure is retained rather than selecting an earlier passing mesh pair or relaxing thresholds. The initial sensitivity writer encountered a NumPy scalar JSON-serialization error; an additive scalar-conversion wrapper produced the report without changing any metric or selected record.

7.2. Appearance and image metrics

The same initial Gaussian appearance is fitted and frozen for both methods; no future images optimize physical predictions or appearance. The image pipeline seals all 2480 declared image slots before reference scoring: 2170 images are scored, and 310 native slots are missing and explicitly retained as failures. On the three paired completed worlds, held-image foreground-union PSNR is 18.734dB versus 12.842dB, foreground-union SSIM is .7473 versus .5770, silhouette IoU is .9562 versus .7950, and jointly visible depth RMSE is 5.478 versus 14.557mm (ours versus native). These are conditional three-world means, not an all-four paired estimate. Shared appearance reduces one source of confounding; different physical trajectories and different transport routines still contribute jointly.

7.3. Development failures and qualified scope

The abrupt 20% compressed release remains spatially unqualified even after its bounded refinement ladder. More aggressive conservative releases and drops can encounter determinant guards; their available prefixes and termination reasons are retained. Energy conservation alone does not guarantee element validity. No artificial restitution or relaxed determinant threshold was introduced to make a failed drop look successful.

A separate smooth position-controlled compression/hold/unload reaches approximately 22.7% actual end-face closure under 30% commanded tool-separation change. Its $n3/n4$ world-coordinate discrepancy is 53.2μm and the declared joint checks pass. A smaller conservative drop passes its $n6/n8$ development checks, whereas that result is not inherited by the fresh material cohort. The viscous 40mm drop and a separate viscous striker pass their timestep pair checks but remain without spatial qualification. The displayed conservative 80g striker is a single-step-size demonstration, without that temporal qualification. Static learned-contact composition has its own qualified sets. These distinct statuses must not be merged into one universal "physics validated" label.

8. Reproducibility, evidence and media

8.1. Source and artifact map

The implementation separates bulk pretraining, interface pretraining, forward mechanics, descriptor inference, and rendering. The following filenames identify the source entry points; they are not additional experimental variants.

- Bulk invariants and basis: materials.py; exact common-scale wrapper: scale_separated_material.py; final pretraining: train_scale_separated_material.py.
- Bilateral law: interfaces.py; final wide-range training: train_interface_range_warm.py; Q2 assembly and differentiable footprints: mechanics.py.
- Normal contact: contact_mechanics.py; dynamics: dynamics.py; separate viscous variant: viscous.py.
- Static inference: joint_inference.py and gauss_newton_inference.py; fresh dynamic inference: cohort_shared_fit.py and matched_material_fit.py.
- Gaussian material transport: gaussian_transport.py; appearance adaptation: cohort_appearance_transport.py; native-author rendering bridge: matched_gaussian_states.py.

The first group resides in the compositional neural-mechanics source package; the dynamic experiment and cohort orchestration reside in the compositional-dynamics source package. Earlier frozen sources and reports are preserved rather than overwritten by recovery implementations.

8.2. Evidence records

Static oracle transfer is summarized in oracle-recovered-plus-stable-io-comparison.json. Its completed analytic/learned oracle artifact recovery passes 3822 independent checks, and the stable-input direct control passes 3570. These checks verify recorded artifacts and calculations, not a new theoretical correctness guarantee.

The unknown-parameter experiment is summarized in unknown-action-reaction-paper-summary.json, with separate action and reaction audits (931 and 1734 checks). The small conditional posterior studies and their recovery histories are distinct from the 15D local uncertainty approximation. Numerical consistency of a small reference posterior does not certify the larger inferred posterior or the dynamic benchmark.

Fresh dynamic scoring is recorded in fresh-cohort-geometry-recovery-results.json. Its independent audit checks 2686 identities, arrays and calculations. Reference sensitivity and post-hoc ranking checks are in fresh-cohort-reference-sensitivity.json and fresh-cohort-reference-robustness-posthoc.json, with 359 and 171 independent checks, respectively. Image results and conditional paired summaries are in fresh-cohort-appearance-results.json and fresh-cohort-appearance-paired-summary.json.

Sealing and recovery. The cohort protocol and source freeze precede private-world generation. Public calibration export binds world identity, protocol identity, observation arrays and metadata. Predictor processes receive public calibration and known forecast setups, not private material labels or future target arrays. Forecast commits include completion or failure and artifact digests. Geometry scoring waits for all original and recovery prediction slots; image scoring additionally waits for prediction-image commitments. Failures remain in cohort denominators. The native environment recovery, direct-baseline IO recovery and scalar serialization wrapper are operational amendments with separate records; none selects a better prediction using held targets.

Budget and precision. Constitutive derivatives and the Q2 mechanics path use float64. Resource limits terminate jobs with explicit incomplete records; optimizer iteration limits and trajectory completion are reported separately. A runtime success flag is not a convergence certificate. Shared law checkpoints are fixed during object fitting. Reproducing the full numerical evidence

requires the pinned sources, checkpoints, canonical geometry, protocol files and local raw trajectory artifacts; the manuscript alone does not replace these inputs.

8.3. Video inventory and interpretation

The final dynamic comparison directory contains all eight files

world-0-forecast-40mm-held0.mp4
world-0-forecast-80mm-held0.mp4
world-1-forecast-40mm-held0.mp4
world-1-forecast-80mm-held0.mp4
world-2-forecast-40mm-held0.mp4
world-2-forecast-80mm-held0.mp4
world-3-forecast-40mm-held0.mp4
world-3-forecast-80mm-held0.mp4

They use synchronized physical times and a fixed held camera. The filename world identifier is opaque and does not encode material labels. Failed native slots are explicitly indicated rather than filled with an apparently successful synthetic motion. The media package contains the twelve MP4 files listed here, a README, and this supplement. Reference and prediction clips display the finite-resolution benchmark, not captured real-world footage.

The four additional development clips are

position-compression-development.mp4
finite-striker-development.mp4
drop-bounce-settle-development.mp4
development-orbit-actual-ours-native-v2.mp4

These development clips are not additional held-cohort successes. Dynamic compression, drop and striker clips use sampled integrated states. No static-contact orbit clip is included in this media package. Slow playback changes presentation time, not the simulated physical timestep. Visual appeal, image fidelity and mechanical accuracy are reported separately.

8.4. Limits of the combined conclusion

The oracle study supports composition of reusable learned laws when physical descriptors are supplied. The unknown-descriptor study shows improved point predictions after an informative additional position observation, while exposing overconfident uncertainty and no advantage over its best fixed probe. The fresh dynamic study supports a conditional improvement over one native adaptation on synthetic load-magnitude extrapolation, with incomplete native coverage and unresolved spatial-reference sensitivity. These findings are complementary, but they do not establish raw-video physical inference, real-world transfer, unknown interaction discovery, frictional dynamics, universally useful active probing, or broad state-of-the-art performance.